

A Study on the Analytical and Visual Comparison of the Mathematical Relationship between CAM and Grad-CAM

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Abstract

With the growing interest in XAI, CAM has been widely used as a representative method to visualize class-specific activation regions in CNNs. Subsequently, Grad-CAM was proposed to overcome the structural limitations of CAM and has become applicable to various networks. This research reexamines the mathematical relationship between CAM and Grad-CAM in general CNN architectures that include GAP, and theoretically reconfirms that when GAP exists, the channel importance coefficient α_k^c of Grad-CAM is proportional to the class-specific weight w_k^c of CAM, as originally presented in the Grad-CAM paper. Furthermore, we empirically validated this by fine-tuning a GAP-based ResNet-50 model on the Stanford Dogs dataset and comparing the heatmaps, bounding boxes, and IoU of the two methods. The results show that in GAP architectures, CAM and Grad-CAM produce virtually identical outputs, clearly demonstrating that the actual behavior of the two methods fully coincides in GAP-based networks.

1. Introduction

In recent deep learning research, convolutional neural networks (CNNs) have demonstrated image classification performance that surpasses human perceptual ability. In the case of ResNet[2], as a deep model, it can autonomously learn features from data and achieve high performance. However, the complex computational structure makes it difficult to explain the rationale behind specific model outputs, leading to the so-called “black-box” problem. Against this backdrop, research in explainable AI (XAI) has emerged to provide visual explanations of model decision-making.

Zhou et al.[8] proposed Class Activation Mapping (CAM), which visualizes regions in an image that the model considers important for classification in the form of heatmaps. This method calculates class-specific activation levels using a global average pooling (GAP)[3] layer immediately before the output. However, applying this approach

requires removing the fully connected (FC) layer and replacing it with GAP, necessitating re-training and introducing structural constraints and practical inconvenience.

To overcome these limitations, Selvaraju et al.[5] introduced Grad-CAM, which utilizes gradient information. Grad-CAM can be applied to models with convolutional layers without modifying the architecture or re-training, and it allows visualization of the activated regions at different layers. Furthermore, Grad-CAM is mathematically shown to be a generalized form of CAM. However, only the mathematical relationship between the two methods was presented, and actual visualization or quantitative comparison was not addressed.

Based on this theoretical background, this research aims to compare and analyze the relationship between CAM and Grad-CAM using a ResNet-based GAP model. The activation maps generated by the two methods for the same model and input images are compared, and bounding boxes are constructed to evaluate qualitative and quantitative agreement with the ground truth using IoU. Through this, we aim to reaffirm that Grad-CAM is a generalized formulation of CAM and to investigate whether the two methods produce identical numerical and visual outputs in GAP-based models.

2. Related Work

Various studies have been conducted to enhance the interpretability of deep learning models. Simonyan et al.[6] proposed the Saliency Map, which highlights the pixels that have the greatest influence on a model’s prediction by computing the gradient of the output class score with respect to the input image. Additionally, Zeiler & Fergus[7] introduced ZFNet, which utilizes deconvolution and unpooling¹ operations to inversely reconstruct the patterns to which CNN neurons respond, thereby improving model interpretability. Building on these approaches, recent research has focused on methods such as CAM and Grad-CAM,

¹Unpooling is an operation that reverses the pooling process to restore spatial information.

which can localize the positions of objects within images.

2.1. Class Activation Map(CAM)

CAM is a method proposed to identify the spatial regions that a CNN focuses on when predicting a specific class. It applies GAP to the last convolutional feature map of the model and generates a class-specific activation map by linearly combining the class-specific weights w_k^c corresponding to each channel.

This approach is simple and intuitive, and it has the advantage of visually indicating the approximate locations of objects that the model has recognized. However, CAM requires the presence of GAP and imposes structural constraints, as it necessitates modifying the architecture of the final classification layer.

2.2. Gradient-weighted CAM(Grad-CAM)

Grad-CAM is a generalized method proposed to overcome the structural limitations of CAM. It computes an importance weight α_k^c for each channel by utilizing the gradient of the class score with respect to the convolutional feature map. This weight reflects how much each channel contributes to the prediction of class c , and a weighted combination of the feature maps is then passed through a ReLU to generate the activation map.

A key advantage of Grad-CAM is that it can be applied directly to existing models without modifying their architecture and can be used for any convolutional layer. Moreover, the Grad-CAM paper demonstrates that it generalizes the mathematical formulation of CAM, suggesting that, in GAP-based models, both methods can produce identical importance weights.

3. Theoretical Analysis

3.1. CAM Formulation

CAM constructs class-specific activation maps by applying GAP to the last convolutional layer feature maps A^k , followed by a linear combination with class-specific weights w_k^c . This formulation provides a direct visualization of the spatial regions used by the model for final object classification.

The GAP operation is defined as:

$$F^k = \frac{1}{Z} \sum_i \sum_j A_{ij}^k, \quad (1)$$

where Z denotes the spatial resolution of the feature map. GAP summarizes the activation level of each channel into a single scalar value, which then serves as the input to the final fully-connected layer of the CNN.

Using the channel-wise average activations obtained by

GAP, the score for class c is computed as:

$$Y^c = \sum_k w_k^c F^k. \quad (2)$$

This represents a linear combination indicating how much each channel contributes to the class prediction.

The final CAM heatmap is constructed as the weighted sum of the feature maps:

$$M_{\text{CAM}}^c = \sum_k w_k^c A^k. \quad (3)$$

This equation illustrates the fundamental principle of CAM, in which convolutional features are linearly combined according to their class-specific importance. Owing to this structure, GAP-based CNNs allow for a relatively clear interpretation of which spatial regions the model has utilized.

3.2. Grad-CAM Formulation

Grad-CAM is an extended technique designed to overcome the structural limitations of CAM, enabling its application to any type of CNN architecture. Unlike CAM, which uses fixed class-specific learned weights w_k^c , Grad-CAM distinguishes itself by directly computing—through gradients—how much each channel contributes when a specific input image produces a specific class score.

To achieve this, Grad-CAM utilizes the gradient of the class score y_c with respect to the final convolutional feature map A_k . These gradients reflect how much each spatial location influences the class score, thereby identifying the important regions.

The importance weight for each channel is defined as:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}. \quad (4)$$

This value quantifies the contribution of each channel to the final decision for class c . Grad-CAM then forms the heatmap by computing a weighted combination of feature maps followed by a ReLU operation:

$$M_{\text{GradCAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right). \quad (5)$$

ReLU is applied to retain only the positive class activations, thereby highlighting the regions that actually support the class decision. Consequently, Grad-CAM operates by estimating channel contributions through gradients and then combining the feature maps accordingly.

3.3. Equivalence of CAM and Grad-CAM in GAP-Based CNNs

Although CAM and Grad-CAM differ conceptually—CAM being weight-based and Grad-CAM being gradient-based—the two methods are analytically equivalent when

the CNN architecture follows a GAP $\rightarrow$ fully-connected structure. In this setting, Grad-CAM naturally reduces to CAM, producing identical heatmaps.

By differentiating CAM's formulation (Eq. (1)–(2)) with respect to A_{ij}^k , we obtain:

$$\frac{\partial Y^c}{\partial A_{ij}^k} = \frac{\partial Y^c}{\partial F^k} \cdot \frac{\partial F^k}{\partial A_{ij}^k} = w_k^c \cdot \frac{1}{Z}. \quad (6)$$

Thus, the gradient at each spatial location of a feature map is directly proportional to the FC weight w_k^c .

Summing this gradient over all spatial locations yields:

$$\sum_{i,j} \frac{\partial Y^c}{\partial A_{ij}^k} = w_k^c. \quad (7)$$

Therefore, the Grad-CAM channel importance becomes:

$$\alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial Y^c}{\partial A_{ij}^k} = \frac{w_k^c}{Z}. \quad (8)$$

Substituting this into the Grad-CAM formulation (Eq. (5)), we obtain:

$$M_{\text{GradCAM}}^c = \text{ReLU} \left(\frac{1}{Z} \sum_k w_k^c A^k \right). \quad (9)$$

Comparing this with the activation map of CAM (Eq. 3), we can see that the only differences between the two methods are the application of ReLU and a simple scaling factor $1/Z$.

This scale difference disappears entirely during visualization because min-max normalization removes it, meaning it has no effect on the final shape of the heatmap. Since ReLU is also commonly applied in CAM during the visualization stage to remove negative activations, this part likewise does not create any substantial visual difference between the two techniques.

As a result, in GAP-based CNNs, CAM and Grad-CAM output the same activated regions. In other words, Grad-CAM is a generalized form of CAM, and when the structural conditions of CAM are satisfied, the two methods theoretically produce the same heatmap.

4. Experiment

4.1. Dataset and Model Architecture

In this experiment, we used a ResNet-50-based classification model with GAP applied. The backbone retained the pre-trained weights from ImageNet[4], and the original fully connected layer was removed and replaced with GAP, followed by a 120-dimensional Dense layer to enable classification on the Stanford Dogs dataset (120 classes) and facilitate CAM extraction.

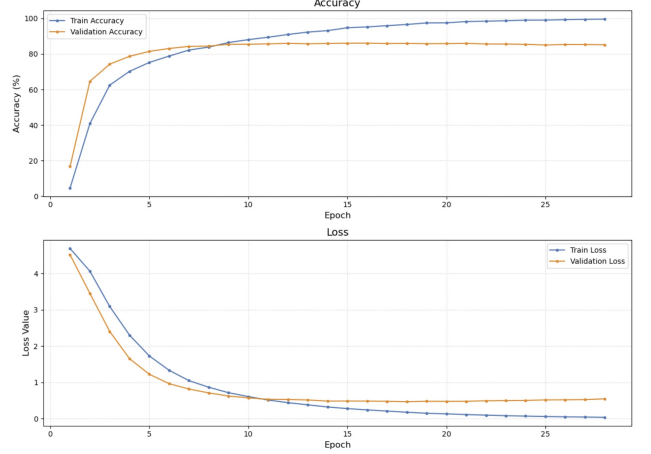


Figure 1. **Overall Training and Validation Curves.** The full learning curve for the Train/Validation set, with the top axis representing Accuracy and the bottom axis representing Loss.

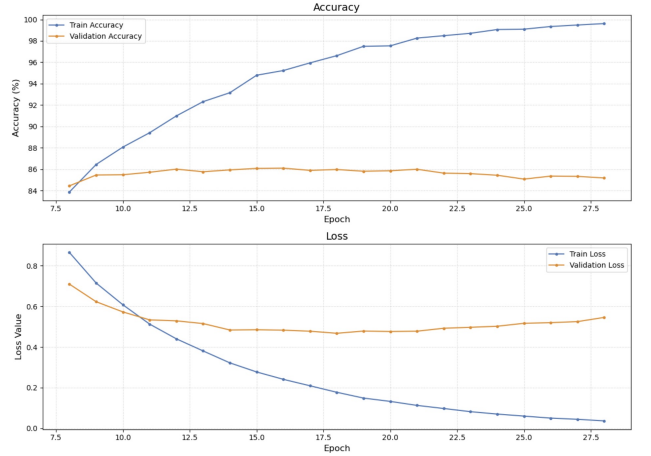


Figure 2. **Zoomed-in Mid-epoch Training Curves.** This learning curve visually emphasizes the mid-to-late stages by zooming in, excluding the initial epoch range.

The dataset used was Stanford Dogs[1], consisting of 12,000 training images and 8,580 validation images, for a total of 20,580 images across 120 dog breeds.

The model was fine-tuned to the dog breed images for a total of 28 epochs. The training process was analyzed using accuracy and loss curves. As shown in Fig. 1, during the early stages of training, the validation accuracy was observed to be higher than the training accuracy. This can be interpreted as the model prioritizing the learning of general visual patterns in the early training stages, before overfitting to specific training samples. However, after approximately 8 epochs, the two curves reversed. training accuracy continued to increase steadily, while validation accuracy plateaued with only minor increases.

A similar pattern was observed in the loss curves. In

the lower graph of Fig. 1, around epoch 10, the training loss decreased more rapidly and crossed the validation loss. Fig. 2 shows a zoomed-in view of the mid-to-late training stage, where training accuracy continued to rise, reaching nearly 99%, whereas validation accuracy slightly declined after epoch 16. The loss curves also show that while training loss kept decreasing, validation loss began to increase again around epoch 18, indicating gradual overfitting in the later stages of training.

4.2. Implementation of CAM and Grad-CAM

Both CAM and Grad-CAM were generated based on the feature maps extracted from the last convolutional block (layer4) of ResNet-50. CAM computes class activation maps by linearly combining the feature maps using the class-specific weights from the fully connected layer immediately following GAP. Although the original CAM formulation does not include an activation function, in this study, negative values were clipped to zero during visualization, leaving only regions that positively contribute to class prediction. This has the same effect as applying ReLU in Grad-CAM mathematically.

Grad-CAM calculates the gradients of the loss with respect to the feature maps for the predicted class, applies GAP to compute the importance of each channel, performs a weighted sum of the feature maps, and applies ReLU to generate the final activation map.

Both methods used the same pre-trained and fine-tuned model, and were computed on the same input images to enable a fair comparison.

4.3. Bounding Box Extraction and IoU Evaluation

The generated class activation maps were first scaled to the $[0,1]$ range using min-max normalization. A fixed threshold of 0.2^2 was then applied to extract a binary mask of regions with activation above this level. The coordinates of all activated pixels in the mask were obtained, and the minimum and maximum values of these coordinates were used to calculate a bounding box. This predicted bounding box was used to evaluate how consistently CAM or Grad-CAM captures the object's location.

Ground-truth boxes were taken from the Stanford Dogs dataset annotations and scaled to match the input resolution (224×224). By visualizing and computing the Intersection over Union (IoU) between the predicted and ground-truth boxes, the object localization performance of CAM and Grad-CAM was quantitatively assessed.

5. Results

Heatmap Comparison. The comparison of CAM and Grad-CAM heatmaps (Fig. 3) showed that the two meth-

²This threshold was chosen arbitrarily.

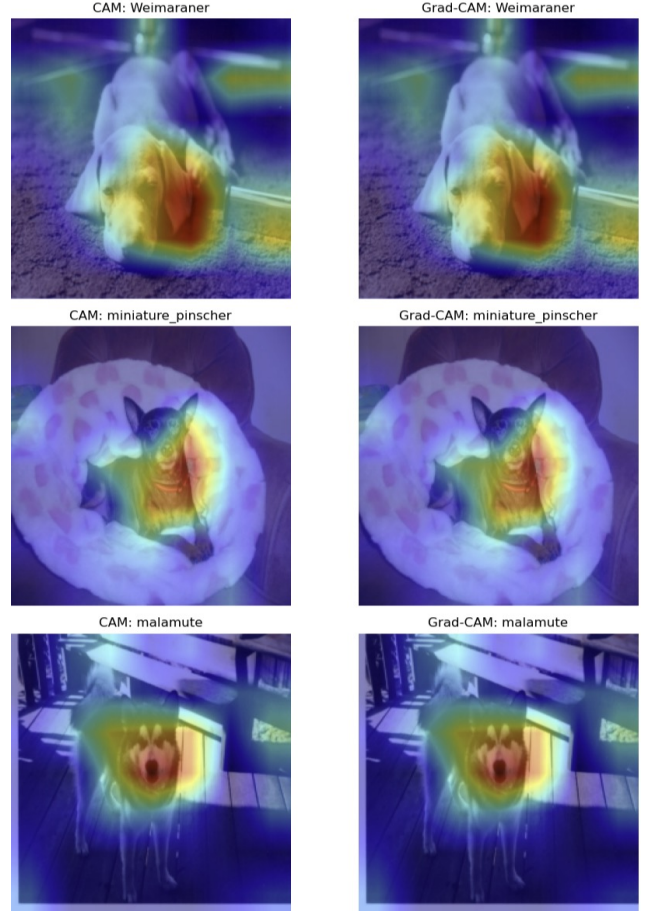


Figure 3. CAM and Grad-CAM Overlay on Original Images. This is the result of interpolating the activation map to match the original image resolution and then overlaying it. The left side shows CAM, the right side shows Grad-CAM, and a total of three example images are presented.

Method	Mean IoU
CAM	0.5695
Grad-CAM	0.5695

Table 1. Comparison of Mean IoU between CAM and Grad-CAM.

ods exhibited identical visual patterns in terms of major activation locations, focus areas, and overall distribution. In particular, both methods produced nearly identical outputs in the key regions where the objects were located, which aligns with the theoretical analysis indicating that GAP-based CNNs inherently use the same channel importance for both methods.

Bounding Box and IoU Comparison. The comparison of bounding boxes (Fig. 4) showed the same trend as the heatmaps. Both methods generated bounding boxes at iden-

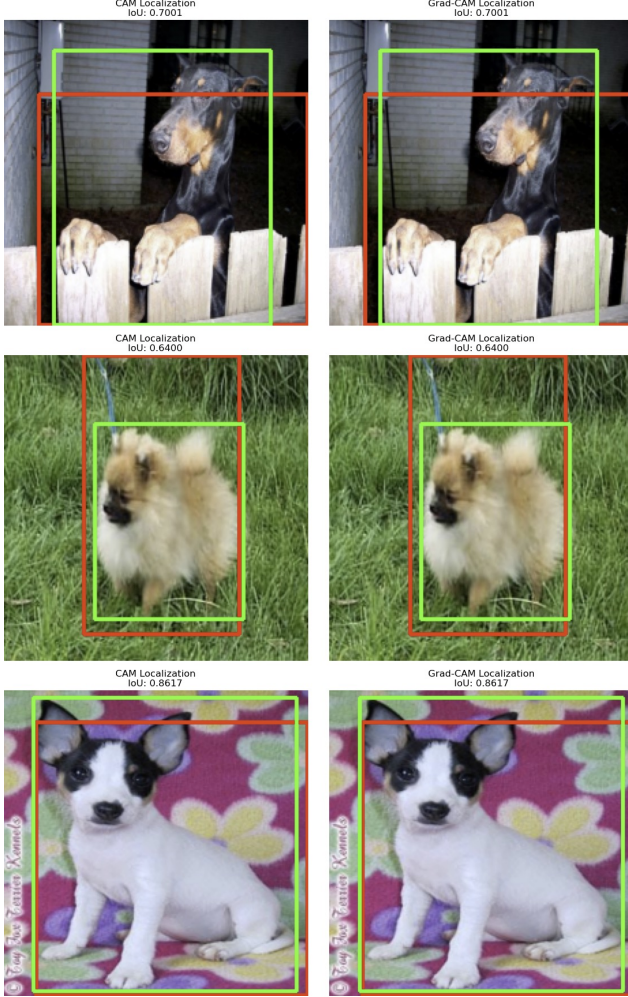


Figure 4. **Bounding Box Comparison between CAM and Grad-CAM.** This is the result of comparing the two methods by representing the activated regions as bounding boxes. The green box is the ground truth (GT) box, the red box is the box predicted by the model, and the IoU value between the two boxes is also marked on each image.

tical locations, and the IoU values for individual validation samples were completely identical in all cases. The mean IoU for the entire validation set, as shown in Table 1, was also the same for CAM and Grad-CAM, both at 0.5695.

These results empirically confirm that the equivalence of channel importance ($w_k^c \propto \alpha_k^c$), as derived mathematically for the two methods, is faithfully reproduced in the actual localization task.

Empirical Verification of Theoretical Equivalence. CAM and Grad-CAM produced identical results across visualizations (heatmaps), localization (bounding boxes), and quantitative metrics (IoU). This directly supports the theoretical analysis that the fundamental structures of the two methods converge to the same form in GAP-based CNNs.

Therefore, while Grad-CAM is a generalized extension of CAM, it is evident that for models satisfying CAM’s structural conditions, both methods provide identical activation regions and equivalent quantitative performance.

6. Future Work

Model Architecture Limitations The present experiments were conducted exclusively on a ResNet-50 architecture explicitly using GAP. In architectures without GAP, CAM cannot be directly applied, making it impossible to empirically verify the equivalence between the two methods. This underscores that Grad-CAM is not merely a reproduction of CAM but a truly generalized method that remains valid in models without GAP.

Visualization Processing In this study, ReLU, min–max normalization ($[0,1]$), and a fixed threshold (0.2) were applied to the CAM and Grad-CAM heatmaps, following the visualization procedure from the original Grad-CAM paper. Furthermore, the scale difference inherent in GAP-based architectures $\alpha_k^c = w_k^c/Z$ was eliminated through normalization, ensuring identical processing for both methods.

These post-processing steps resulted in 100% agreement in heatmaps, bounding boxes, and IoU across all validation samples. Omitting min–max normalization or ReLU could introduce minor numerical differences, but these would reflect only differences in visualization pipelines; mathematically, as demonstrated in Section 3.3, the two methods remain equivalent.

In conclusion, the current pipeline represents an optimal setting to empirically validate the claim that CAM = Grad-CAM. Future research could further explore the equivalence under various post-processing modifications to reinforce this finding.

7. Conclusion

In this study, we re-derived the mathematical relationship between CAM and Grad-CAM using a GAP-based ResNet-50 model and compared the two methods’ heatmaps, bounding boxes, and IoU on the Stanford Dogs dataset. The experimental results showed that both methods produced virtually identical activation regions and quantitative performance in the GAP-based model, empirically confirming that Grad-CAM is a generalized form of CAM.

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